

AGASTYA SHARMA

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TECHNICAL SKILLS

ML / Deep Learning: PyTorch, TensorFlow, Transformers, Computer Vision, Model Evaluation, Quantization, QLoRA

LLMs / Generative AI: RAG, Multi-Agent Systems, Anthropic SDK, OpenAI API, Gemini API, MCP, Langfuse

Languages: Python, C, C#, SQL, TypeScript

ML Infrastructure & Cloud: Hugging Face, GGUF, llama.cpp, Docker, AWS (S3, Bedrock), Azure, PostgreSQL, Redis

Frameworks: React, Next.js, Node.js, REST APIs, GraphQL, .NET / EF Core

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science, Co-op

2025 – Present

- President's Scholarship of Distinction; Michael Egan Entrance Scholarship in Computer Science

WORK EXPERIENCE

Intercept

Feb 2026 – Present

Solutions Engineering Intern (part-time)

Toronto, ON

- Built a distributed TypeScript/Node.js **Generative Engine Optimization** pipeline running **347K+ LLM inference jobs** across **6 AI providers** for 3 enterprise clients, generating **+\$30K in ROI**
- Architected an end-to-end **multi-agent pipeline** on the **Anthropic SDK** with agent orchestration, **SharePoint-grounded RAG**, and **Langfuse**-based evaluation of LLM outputs, cutting content generation time by 80%
- Built a reusable **MCP server** that served enterprise SharePoint context to LLM clients
- Wrote AI architecture documentation adopted by a **20+ developer** team as the standard for extending and maintaining AI systems

Sylphia Consulting Inc.

Sep 2025 – Feb 2026

Full-Stack Software Developer (part-time)

Toronto, ON

- Developed **Python computer vision** pipelines using **PyTorch** depth estimation models for image recognition
- Deployed a cross-platform React Native app on **AWS Bedrock** and S3 for users across India; shipped the Next.js/TypeScript app in 4 months, reaching **1,000 users** and raising \$10K in funding

University of Waterloo

May 2026 – Aug 2026

Junior Application Developer (co-op)

Waterloo, ON

- Built and deployed **4 production pages** for an academic-review platform (React/.NET); authored **55 Jest tests** covering role-based access, routing, and file uploads, raising critical-flow coverage by 23%
- Learned .NET/EF Core in an unfamiliar codebase, independently diagnosing an API-layer migration issue and adding 8 regression tests to prevent recurring data-integrity failures

PROJECTS

Tiny Aya Quantization Safety Audit | *llama.cpp, PyTorch, QLoRA, Hugging Face, GGUF*

Sep 2026

- Audited whether Tiny Aya's multilingual safety survives shipped **GGUF** quantizations (Q8_0, Q4_K_M, Q4_0) against the BF16 baseline using **MultiJail** across 10 languages, targeting a gap in the paper, which evaluates quantization only on generation quality
- Built a blocking **backend-parity gate** (prompt token-ID equality, stop-token checks) so measured deltas reflect quantization, not the **llama.cpp** vs. Transformers harness; added a **QLoRA** safety-repair pass with GGUF re-export and re-audit

V-JEPA Robotics Dataset Pipeline | *Python*

Mar 2026

- Built an **8-stage** data curation pipeline for V-JEPA-style, action-conditioned world models, turning raw robot logs into training-ready windows with **clock-drift correction**, temporal alignment, gap-aware sampling, **9 quality flags**, and **17D state vectors**

Memento – Memory Support for Dementia Patients | *Next.js, Flask, InsightFace, Gemini*

Jan 2026

- Built real-time **face recognition** (InsightFace, facial embeddings) and **speaker diarization** with voice transcription feeding a caregiver-facing Gemini assistant, at DeltaHacks 2026

parsemd – Document Context Plugin for Claude Code

Aug 2026

- Built a Claude Code plugin for document/audio context injection with **token budgeting**, caching, page slicing, and sandboxed execution